

CONEPRO · DESIGN PRECIS

ConePro design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

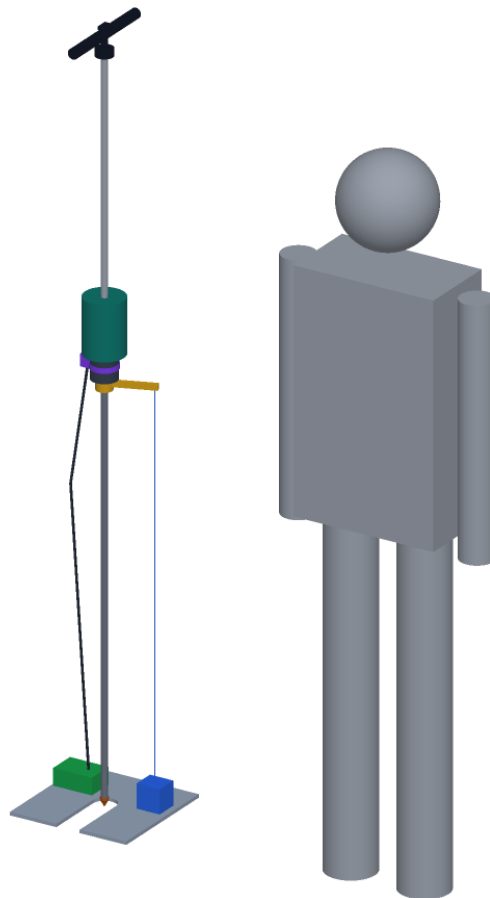
Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update; design choices decided (CNP-DDR-001); numbers replaced by CNP-CAL-001 results; parametric model, drawing CNP-DWG-001 and refreshed media
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

ConePro design precis

ConePro is a standard-geometry dynamic cone penetrometer (8 kg hammer, 575 mm drop, 20 mm 60 degree cone, as in ASTM D6951) with a draw-wire depth sensor referenced to a plate on the ground, a clamp-on sensor pad at the anvil that detects blows and reads rod tilt, and a small BLE logger, so one person can run a test while the phone plots penetration against blows and exports the record. The sensor set is designed to fit existing standard penetrometers too. The TRL 3 calculations (CNP-CAL-001) give about 28 J at the cone per blow, 850 mm of usable penetration per lower rod, about 40 h of logging on three AA cells and \$321 in parts, within the \$400 budget. Two requirements are not met: the instrument carried in its bag weighs about 16.7 kg against 16 kg (R10), and there is no extension rod for the 1,000 mm clause of R5. Depth accuracy (R2) and shock survival (R11) are at risk.

ConePro

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale

Figure 1. ConePro set up at the start of a test, with a 1.75 m person for scale. The cone tip rests on the ground through the slotted reference plate and the hammer rests on the anvil.

■ How it works

1. **Set up.** The operator lays the slotted reference plate on the ground, stands the cone in the plate's center hole, clamps the draw-wire arm to the rod just below the anvil and straps the sensor pad to the anvil. The app zeroes the depth with the cone point at the surface, and zeroes the tilt by turning the rod half a turn.

2. **Drive.** The operator holds the handle, lifts the 8 kg hammer to the top stop and lets it fall 575 mm onto the anvil. Each blow drives the rod and cone further into the soil.
3. **Sense.** A ring of magnets in the hammer face passes a potted Hall-effect switch in the sensor pad as the hammer lands, and the pad's accelerometer sees the impact peak. A blow counts only when both fire and the depth steps down or holds within 0.5 s. Between blows the accelerometer reads rod tilt. A draw-wire sensor on the plate measures how far the clamp arm has moved down, which equals the cone's penetration because the rod is rigid and the plate stays on the ground surface.
4. **Log.** The logger on the plate reads depth at 100 Hz, waits 0.2 s for the rod and wire to settle after each blow, and stores blow number, depth, tilt and time. It keeps the record even when no phone is connected.
5. **Show.** The phone app plots depth against blows, corrects depth for rod lean, splits the curve into layers of near-constant slope, and gives each layer's DCP index (mm per blow) and an indicative CBR from the ASTM D6951 correlations, with the correlation named on screen. It warns when the rod leans more than 5 degrees. The user exports a CSV with time, location and notes.
6. **Extract.** After the test the operator removes the plate over its slot and pulls the rod out with the optional lever rod puller, which is carried separately.

ConePro: energy per blow, J (estimates)

CONCEPT, NOT FOR FABRICATION

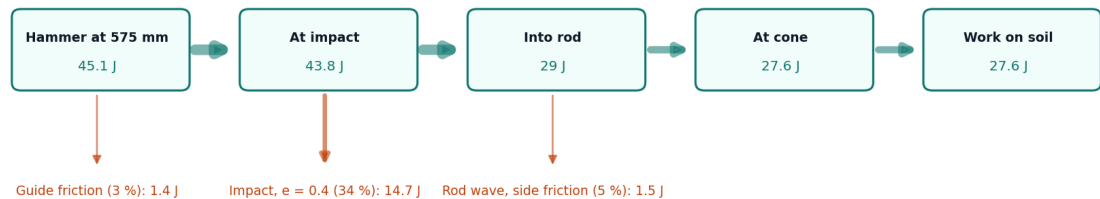


Figure 2. Energy per blow from hammer to soil, in joules (estimates from CNP-CAL-001, section B). The impact loss depends on the restitution between hammer and anvil, which is assumed, not measured.

■ Main components

Numbers match the exploded view (Figure 3) and `bom/bom.csv`. Geometry is in the parametric model `cad/src/model.py` and on drawing CNP-DWG-001.

#	Component	Choice	Notes
1	Cone	Hardened steel, 20 mm base, 60 degree point, threaded onto the rod	Replaceable; a worn cone biases results
2	Lower drive rod	16 mm steel, 1,000 mm, engraved every 10 mm	Engraving is a manual backup scale and the calibration reference
3	Anvil and coupler	Steel, 64 mm diameter x 60 mm, threaded between the rods	Plain; the sensor pad clamps on (D3)
4	Drop hammer	8.0 kg steel, 100 mm OD x 22 mm bore x 136.4 mm	Sixteen 8 x 4 mm magnets in the lower face
5	Upper rod (hammer guide)	16 mm steel, 729 mm	Gives the 575 mm free drop to the stop
6	Handle and top stop	26 mm tube T-handle, 44 mm stop collar, bubble level	Level is the tilt backup (D1)
7	Reference plate	300 x 300 x 8 mm aluminum, 60 mm center hole and slot	Depth reference on the ground surface
8	Draw-wire depth sensor	60 mm grooved aluminum drum, single layer, constant-force spring, 12-bit angle sensor, groove keeper, preloaded eye spring	Drum must be metal for thermal stability; bought industrial sensor is the fallback
9	Anvil clamp and wire arm	Split collar on the lower rod, arm about 115 mm to the wire eye	Wire exit on the plate 105 mm from the rod
10	Blow and tilt sensor pad	Hall switch and 3-axis accelerometer, potted, on a 300 Hz elastomer isolator, band clamp for 50 to 80 mm anvils	Top 8 mm below the anvil top, under the hammer overhang
11	BLE logger	Microcontroller with BLE, 4 MB flash, button and LED, three AA cells, IP65 box	Kept on the plate, away from impacts
12	Coiled sensor cable	Pad to logger, with a strain relief at each end	Follows the rod down during the test
15	Rod extraction lever (optional)	Lever puller with a self-gripping 16 mm rod clamp, about 10:1, about 3 kg	Carried separately; outside the R13 total (D2)

ConePro: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Hardened cone, 20 mm, 60 degree
- 2 Lower drive rod, 16 mm x 1,000 mm
- 3 Anvil and coupler
- 4 Drop hammer, 8 kg
- 5 Upper rod (hammer guide)
- 6 Handle, top stop, bubble level
- 7 Reference plate, slotted
- 8 Draw-wire depth sensor
- 9 Anvil clamp and wire arm
- 10 Blow and tilt sensor pad
- 11 BLE logger, 3 x AA
- 12 Coiled sensor cable

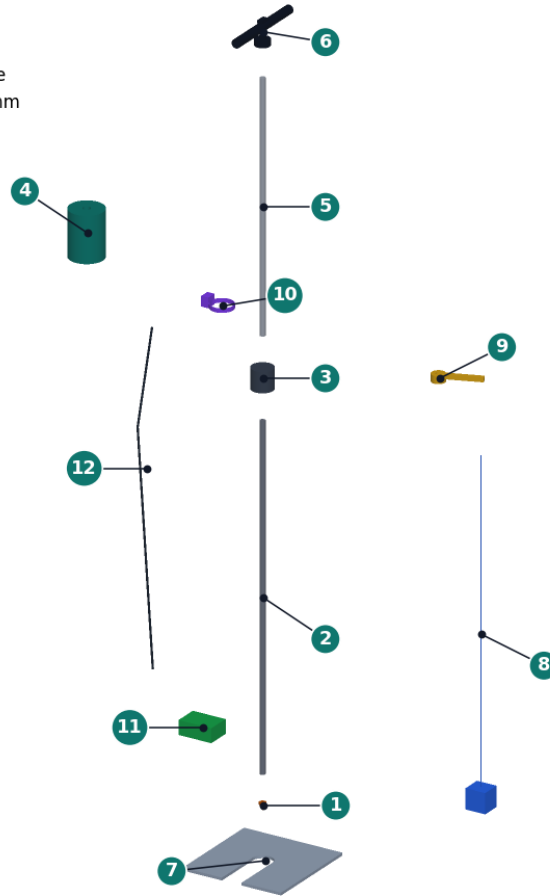


Figure 3. Exploded view with numbered callouts matching the BOM. The optional extraction lever is not shown.

■ Key numbers

All values are estimates from CNP-CAL-001; the bracketed tag names the line of `docs/04-calcs/sizing.py` that prints each one.

Table 1. Blow energy and timing.

Quantity	Estimate	Basis
Potential energy per blow	45.1 J [B1]	8.0 kg x 9.81 m/s ² x 0.575 m
Impact speed; fall time	3.36 m/s; 0.342 s [B1]	Free fall over 575 mm
Energy reaching the cone	27.6 J, 61 % (25.6 to 30.9 J) [B4]	3 % guide friction; Hiley impact efficiency with restitution 0.4 (0.2 to 0.6) and 5.35 kg driven mass; 5 % rod and side friction
Mean dynamic soil resistance	1.8 kN at 15 mm per blow; 13.8 kN at 2 mm per blow [B6]	Energy at the cone over penetration
Blow rate	17 to 24 per minute	One blow every 2.5 to 3.5 s (assumed)

What the numbers mean for the user. The ASTM D6951 correlation for most soils is $CBR = 292 / DCP^{1.12}$, with DCP in mm per blow (separate correlations apply to lean clays below CBR 10 and to fat clays). Table 2 shows how sharply the indicative CBR changes at small penetrations, which is why depth accuracy matters most in stiff layers.

Table 2. Indicative CBR from the ASTM D6951 general correlation [E7].

DCP index (mm per blow)	2	5	10	15	25	50
Indicative CBR (%)	134	48	22	14	8	4

Table 3. Depth measurement.

Quantity	Estimate	Requirement
Resolution	0.046 mm per count [E1]	R2 resolution met
Bench error over 1,000 mm, vertical rod	±0.98 mm worst case; ±0.49 mm root sum square [E3]	R2 (±1 mm) at risk
Extra error from rod lean at 850 mm, after tilt correction	0.96 mm at 1 degree; 1.81 mm at 2 degrees [E4]	Why the usable range stops at 850 mm
DCP index error, 10 blows at 2 mm per blow	±4.9 % RSS (±9.8 % worst); CBR ±5.5 % [E8]	
Wire slack after a blow	1.9 to 17 mm for 5 to 36 ms [E5]	Groove keeper and preloaded eye spring needed
Record latency	0.21 s after impact [E6]	R4 met
Plate settlement or heave	Not calculable; static pressure only 0.30 kPa [E9]	Open question

Table 4. Range, mass, time, shock and power.

Quantity	Estimate	Requirement
Stroke per lower rod	938 mm, limited by the wire arm meeting the reel; 850 mm usable [D1, D2]	R5 850 mm met; 1,000 mm clause not met (no extension rod)
Assembled height	1,867 mm [A3]	
Mass as carried in the bag	16.7 kg; 15.9 kg without the bag [C2]	R10 (16 kg) not met
Heaviest piece	9.82 kg, hammer captive on the upper rod [C3]	R10 (10 kg) met
Packed length	About 1.08 m (longest piece 1,022 mm) [C4]	R10 met
Reference test time	8.1 to 11.7 min [J1]	R9 met, thin
Shock at the anvil; at the isolated pad	1,400 to 5,400 g mean; about 533 g [G1, G2]	R11 at risk
Tilt reading	±0.3 degree after rotation zeroing [H1]	R6 met by design
Average current; battery life	50.2 mA; 40 h, 20 h at 0 °C [K1, K2]	R12 met
Storage	About 830 tests in 4 MB [I1]	R8 met

Table 5. Cost [L1].

Group	Indicative cost
Mechanical DCP (items 1 to 7)	\$174
Sensing and logging (items 8 to 12)	\$102
Carry bag, hardware and consumables (items 13, 14)	\$45
Instrument total	\$321 (R13, \$400, met)
Optional extraction lever (item 15)	\$50, outside the instrument total

■ Key design choices

Amish decided the TRL 2 review items on 2026-09-25 by accepting every recommendation (CNP-DDR-001). The choices below are therefore decided, not proposed.

- **ASTM D6951 geometry (D5).** The standard 8 kg hammer and 575 mm drop, so users can apply published correlations and compare with existing DCP data. The 4.6 kg option is noted for later.
- **Depth by draw-wire from a ground plate (D4).** A draw-wire reel on the plate measures the clamp arm's movement. A laser time-of-flight sensor is the fallback to compare later. TRL 3 adds three details to meet the calculations: a metal, single-layer grooved drum; a groove keeper for the slack after each blow; and a preloaded spring at the arm eye to limit the snap tension.
- **Blow detection by Hall sensor with cross-checks (D6).** A blow counts when the Hall switch sees the hammer arrive, the accelerometer sees an impact peak and the depth steps down or holds within 0.5 s. A peak below half the full-drop value flags a short drop.
- **Tilt by accelerometer, with a bubble level as backup (D1).** The accelerometer in the sensor pad reads tilt between blows, and the app warns above 5 degrees and corrects depth for lean.

- **Electronics on the plate (D7).** Only the isolated sensor pad sees impacts.
- **Three AA cells (D8).** No lithium safety or shipping issues; sold everywhere.
- **Optional lever for extraction (D2).** A lever puller with a rod clamp, carried separately and outside the instrument cost and mass.
- **Sensor set fits standard DCPs (D3).** The clamp, band-mounted pad, plate and reel need no machining of the penetrometer (R15). The prototype is a complete instrument.

The TRL 3 calculations raised new items that need Amish's decision (R10 mass, the R5 extension rod, and the reel details above). They are listed in `docs/REVIEW.md` as proposed, awaiting Amish.

■ Safety

Safety: ConePro drops an 8 kg steel hammer onto a steel anvil every few seconds and drives a pointed steel rod into the ground. Treat buried services, crushed fingers, flying fragments, noise, back strain and misuse of the results as hazards at every test point.

- **Buried services.** A rod driven 1 m into the ground can strike electric cables, gas pipes and water mains. Get a utility locate before every test (in the United States, call 811 or use the state one-call service) and do not test within the marked tolerance zone.
- **Crushed fingers and hands.** The hammer lands with about 44 J. Keep hands on the handle only, never on the anvil, the hammer's lower face, the sensor pad or the clamp arm. The pad sits 8 mm under the hammer overhang, which is a pinch point: fit and remove it only with the hammer held at the top stop or removed.
- **Flying fragments and mushrooming.** Repeated steel-on-steel impact at thousands of g can chip the anvil or hammer face and spall hardened steel. Wear safety glasses, inspect the faces, and replace them when they mushroom or crack.
- **Magnets.** The hammer carries sixteen strong magnets. Keep it away from pacemakers and other implanted devices, and from cards and instruments that magnets can damage.
- **Draw-wire.** A wire that breaks or is released under tension can whip. Keep faces away from the reel and never let the arm snap back.
- **Noise.** Steel-on-steel impact is loud at close range. Hearing protection is recommended; the level is not yet estimated.
- **Lifting and posture.** The hammer is lifted about 57 times per test and the upper assembly weighs about 9.8 kg. Use a straight back and swap operators on long test days. Pull the rod only with the lever, never by back strength; stop if the lever will not move it (the pull force can exceed 3 kN in stiff ground).
- **Rod whip and tipping.** A leaning rod can bend or kick sideways when struck. Stop when the app warns of lean (R6).
- **Sharp cone.** Cover the cone in transport.
- **Ground and site hazards.** Contaminated land, trench edges and traffic: follow site rules and do not test at the edge of an open excavation.
- **Misuse of results.** CBR values are indicative correlations. ConePro results must not be used on their own to design foundations or to decide that ground is safe to build on; they show where a qualified geotechnical engineer should investigate.
- **Batteries.** Alkaline AA cells carry little risk; remove them for storage to avoid leakage.

■ Open questions

- Choose the first co-design partner and site types (proposed, awaiting Amish; partners to be picked per area later).
- Decide the TRL 3 proposals in `docs/REVIEW.md` : R10 mass reduction, the R5 extension rod, and the reel details.
- Measure the restitution, the magnet field with the steel hammer, the isolator response and the draw-wire error on a bench.
- Estimate plate settlement or heave under blows, and how to detect it (for example a second reference point).
- Define the blow and depth data format and the layer-splitting method; check against ORN 8 practice.
- Confirm commercial prices for instrumented penetrometers (not yet checked online).

Concept media: [blueprint sheet](#), [interactive 3D model](#). General arrangement: [CNP-DWG-001](#).